



North South University

Department of Electrical & Computer Engineering

Assignment

Course Code: CSE 425

Course Title: Concepts of Programming Language

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Submission Date:09-08-2025

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A, C program to tokenize of a C statement that can identify the lexeme with its category and can find out the keyword of C language ---

Source Code:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

const char *keywords[] = {
    "auto", "break", "case", "char", "const", "continue", "default", "do",
    "double", "else", "enum", "extern", "float", "for", "goto", "if",
    "inline", "int", "long", "register", "restrict", "return", "short",
    "signed", "sizeof", "static", "struct", "switch", "typedef", "union",
    "unsigned", "void", "volatile", "while", "_Alignas", "_Alignof",
    "_Atomic",
    "_Bool", "_Complex", "_Generic", "_Imaginary", "_Noreturn",
    "_Static_assert", "_Thread_local"
};

int keywordCount = sizeof(keywords) / sizeof(keywords[0]);

int isKeyword(const char *str) {
    for (int i = 0; i < keywordCount; i++) {
        if (strcmp(str, keywords[i]) == 0)
            return 1;
    }
    return 0;
}

int isNumber(const char *str) {
    for (int i = 0; str[i] != '\0'; i++) {
        if (!isdigit((unsigned char)str[i]))
            return 0;
    }
    return strlen(str) > 0;
}

int isIdentifier(const char *str) {
    if (!isalpha((unsigned char)str[0]) && str[0] != '_')
        return 0;
    for (int i = 1; str[i] != '\0'; i++) {
        if (!isalnum((unsigned char)str[i]) && str[i] != '_')
            return 0;
    }
}
```

```

        return 1;
    }

int isOperator(char c) {
    const char *ops = "+-*/%=<>&!^~";
    return strchr(ops, c) != NULL;
}

int isPunctuation(char c) {
    const char *punc = ";,(){}[]";
    return strchr(punc, c) != NULL;
}

int main() {
    char line[200];
    printf("Enter a C statement: ");
    fgets(line, sizeof(line), stdin);

    printf("\nTokenizing...\n\n");

    char token[100] = "";
    int pos = 0;

    for (int i = 0; line[i] != '\0'; i++) {
        char c = line[i];

        if (isspace((unsigned char)c) || isOperator(c) ||
isPunctuation(c)) {
            if (pos > 0) {
                token[pos] = '\0'; // end string
                if (isKeyword(token))
                    printf("%s --> Keyword\n", token);
                else if (isNumber(token))
                    printf("%s --> Number\n", token);
                else if (isIdentifier(token))
                    printf("%s --> Identifier\n", token);
                else
                    printf("%s --> Unknown\n", token);
                pos = 0;
            }

            if (isOperator(c))
                printf("%c --> Operator\n", c);
            else if (isPunctuation(c))
                printf("%c --> Punctuation\n", c);
        } else {

```

```

        token[pos++] = c;
    }
}

if (pos > 0) {
    token[pos] = '\\0';
    if (isKeyword(token))
        printf("%s --> Keyword\\n", token);
    else if (isNumber(token))
        printf("%s --> Number\\n", token);
    else if (isIdentifier(token))
        printf("%s --> Identifier\\n", token);
    else
        printf("%s --> Unknown\\n", token);
}

return 0;
}

```

Output snippet:

```

"E:\summer25\CSE425 Section 6\assignment\tokenizer.exe"
Enter a C statement: float average = (x + y) / 2;

Tokenizing...

float --> Keyword
average --> Identifier
= --> Operator
( --> Punctuation
x --> Identifier
+ --> Operator
y --> Identifier
) --> Punctuation
/ --> Operator
2 --> Number
; --> Punctuation

Process returned 0 (0x0)   execution time : 34.673 s
Press any key to continue.

```